Image, word and thought: A more challenging language task for the Iterated Learning model

Hyoyeon Lee[1], Seth Bullock[1], and Conor Houghton[*1]

[*]Corresponding Author: conor.houghton@bristol.ac.uk
[1]Faculty of Science and Engineering, University of Bristol, England

The iterated learning model simulates the transmission of language from generation to generation in order to explore how the constraints imposed by language transmission facilitate the emergence of language structure. Despite each learner starting from a blank slate, the presence of a learning bottleneck can lead to the emergence of structured language. Such languages lack ambiguity, are governed by grammatical rules, and remain consistent over successive generations; that is, they become expressive, compositional, and stable. Here, for the first time, the semi-supervised iterated learning model has been applied to a language learning task involving the communication of complex visual meanings: noisy seven-segment display images. Agents in this model are able to learn and transmit a language that is expressive: distinct codes are employed for all 128 glyphs; compositional: signal components consistently map to meaning components, and stable: the language does not change from generation to generation.

1. Introduction

Synthetic and computational approaches to understanding the basic properties of language change are attractive, since they offer an additional window on evolutionary linguistic phenomena (Christiansen & Kirby, 2003; Kirby, 2002). In particular, computational agent-based models of language learning can offer insight into the way language-learning constraints might influence, shape, or determine language properties (Kirby, 2002). One such model, the iterated learning model (ILM) (Kirby & Hurford, 2002), has been successfully applied to simple artificial language learning tasks in which meanings and signals are typically limited to short bit strings. Here, this model is extended to a more complex task in which the meanings are glyphs based on the seven-segment LED displays that are often used on calculators and digital watches (Clark, 1929).

To our knowledge, this is the first time that an ILM has been demonstrated on a task this complex. The *semi-supervised ILM* is used. This version of the ILM includes a cognitive scheme that requires agents to employ their language internally as a kind of "language of thought". We believe that this mimics an important step in the evolution of cognition in which the internalization of language allows the sort of chain-of-thought reasoning that has also become important in machine learning (Wei et al., 2022). Extending the model from bit strings to glyphs in-

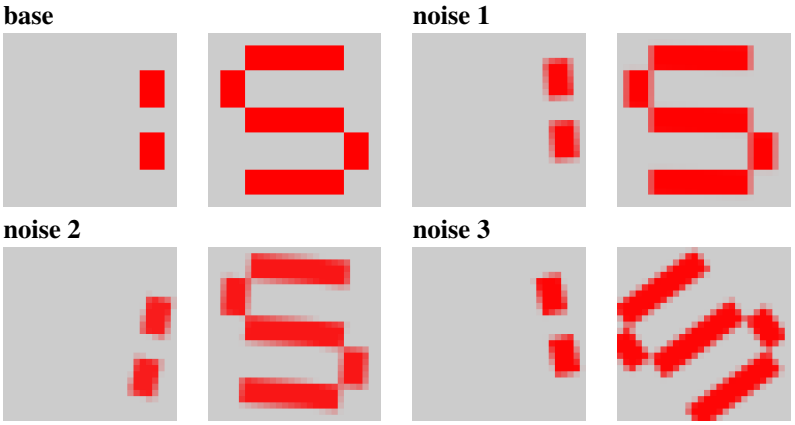

Figure 1. **Example images.** Each base glyph is a 28×28 image based on the classic seven-segment LED display. All 128 glyphs are employed, many of these do not correspond to digits or letters. The two glyph types presented here correspond to digits '1' and '5'. For each base glyph, 100 variant images were produced using small random rotations, translations and changes in intensity; three levels of noise were used, labelled noise 1, noise 2 and noise 3.

volved an extra training step for image encoding and decoding so there is a segregation in learning between low-level representations of stimuli and high-level learning based on those representations.

2. Method

The ILM is an agent-based simulation designed to explore conditions under which languages tend to become expressive, compositional, and stable (Kirby, 2001; Kirby & Hurford, 2002). An ILM comprises a tutor agent and an initially naïve pupil. The tutor exposes the pupil to a finite set of utterances, each pairing a meaning with the signal that expresses it. After this period of instruction, the pupil becomes a tutor and teaches a new initially naïve pupil in the same fashion. A language learning bottleneck is a key feature of the model: since each pupil is only exposed to a finite subset of the tutor's language during learning, when it becomes a tutor itself it will need to generate signals for meanings that were not part of its learning experience. This demand for generalization is critical to the behaviour of the ILM (Kirby & Hurford, 2002).

The ILM introduced by Kirby and Hurford (2002) is a very simple abstract model of language learning: the 'language' is an Encoder map, E, from bit-string meanings, $\mathcal{M}$ to bit-string signals, $\mathcal{S}$. The agent also has a Decoder map D mapping signals to meanings.

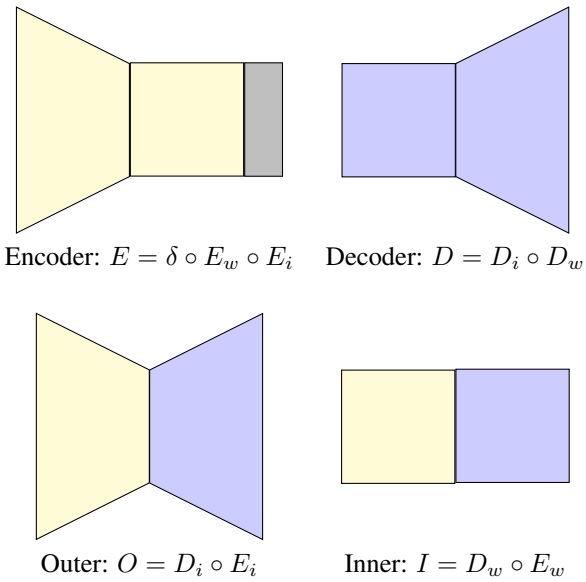

Figure 2. **The network** is trained in four different configurations: as a supervised Encoder mapping glyphs to signals, the grey rectangle denotes the discretization to zeros and ones; as a supervised Decoder, mapping words or thoughts to glyphs; and as two types of unsupervised autoencoder: the Outer maps a glyph to itself, representing the training of the primary visual system, while the Inner maps through the inner parts of the network and represents the internalization of language.

$$\begin{tikzcd} \mathcal{S} \arrow[rr, bend right, "D"'] & & \mathcal{M} \arrow[ll, bend right, "E"'] \end{tikzcd}$$

Three properties of these languages are studied, expressivity, compositionality, and stability; these properties are chosen because they are properties of real languages which can be mapped to quantifable properties of these simple model languages. The expressivity, compositionality, and stability of the languages were quantified using three measures x, c and s. Each of these ranges from zero, for a random language, to one, for a language that is completely expressive, compositional, or stable. The definitions for these measures are given in an extended version of the paper[1].

In the more complex example present here, meanings are 128 seven-segment glyphs, each a 28×28 image subjected to random rotations, translations and in-

[1]A longer version of this paper is available on arXiv as `arXiv:2601.02911`. The glyphs are available at `github.com/IteratedLM/2025_12_7Seg_data`; the ILM code and results data are available at `github.com/IteratedLM/2025_05_7Seg`

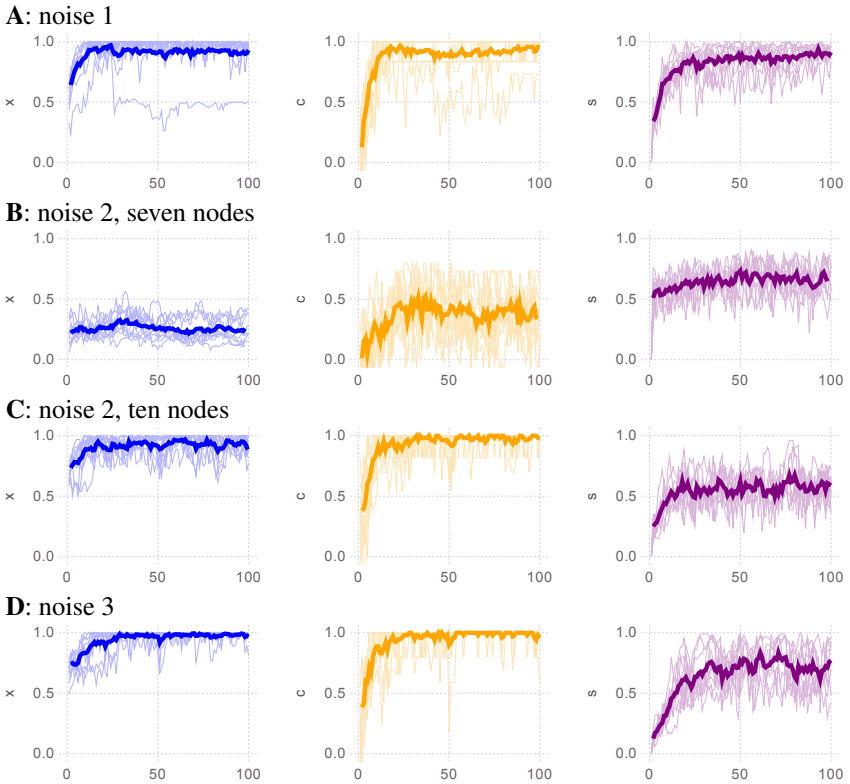

Figure 3. **An expressive, compositional, and stable language evolves**. Thin lines show measures of expressivity, x, compositionality, c, and stability, s, for ten individual 100-generation instantiations. Thick lines show the average values. **A** uses the noise 1 condition with $n_\ell = 7$ latent nodes. In **B** the language does not evolve to be fully compositional or expressive if there are $n_\ell = 7$ latent nodes. However, expanding to $n_\ell = 10$ means a compositional and expressive language evolves. With a large enough number of latent nodes the model can achieve an expressive and compositional language even for noise 3 images: **D** show results where E_w is $20 \times 18 \times 15$.

tensity changes, Fig. 1, and the signals communicating them are bit strings. Following Bunyan, Bullock, and Houghton (2025) both E and D are feed-forward neural networks. In the experiments reported here different sizes of latent space, n_ℓ, and different shapes for the encoder will be used. However, in the simplest case $n_\ell = 7$, and the interior layers all have the same size. For clarity, this case is used in the following descriptions. The encoder is a $784 \times 128 \times 7 \times 7 \times 7$ all-to-all network mapping from the glyphs to latent space vectors, these can be considered to represent an agent's 'thoughts'. It is useful to consider the encoder as comprising two parts, the image encoder E_i, the $784 \times 128 \times 7$ network that reduces the dimension of the image, and the word encoder E_w, the $7 \times 7 \times 7$ map

from the reduced representations to the latent space. The latent space activations are mapped to 'words' by a discretization map δ. The decoder maps are similar, but map from the latent space back to the glyphs.

Crucially, for training to succeed, the neural networks need to be trained in different configurations: as Encoder $E = E_w \circ E_i$ and Decoder $D = D_i \circ D_w$, using examples provided by the tutor, and as two types of autoencoder, the Outer autoencoder $D_i \circ E_i$ which maps the glyphs down to a seven-dimensional representation and then back to the original glyph, and the Inner autoencoder $D_w \circ E_w$ that maps the reduced representations to the latent space and then back to a reduced representation, Fig. 2. The details of how the models are trained are available in the extended version of the paper.

The ILM is a highly simplified abstraction so any claim of correspondence between aspects of the model and the situation in the real world is intended only to point to areas where a metaphorical recreation of the real-world situation improves the dynamics of the model and to suggest these as areas to study further using more detailed models or real-world experiments. In this context it is interesting to suppose the two autoencoders represent training of the primary sensory pathways and of an internalized language (Meister, Wong, Baylor, & Shatz, 1991; Shatz & Stryker, 1978; Cang & Feldheim, 2013; Vygotsky, 1986; Carruthers, 2002; Bunyan et al., 2025; Bullock & Houghton, 2024).

3. Results

In our simulations, the language becomes expressive, compositional, and stable. Fig. 3**A** plots the course of evolution for $n_\ell = 7$ using the noise 1 glyphs. x, c and s quickly approach one. The model also succeeds in evolving an expressive and compositional language for the noisier datasets; however, this requires a higher value of n_ℓ. In Fig. 3**B** a network with $n_\ell = 7$ latent nodes fails to cope with the noise 2 dataset: x and c generally remain below 0.5. However, increasing the number of latent nodes to $n_\ell = 10$ allows an expressive and compositional language to develop quickly, Fig. 3**C**. With a large enough network, an expressive and compositional language can also be achieved for noise 3 images, Fig. 3**D**.

In these examples, the language never fully stabilizes. With larger latent spaces, $n_\ell > 7$, compositionality appears in only seven signal bits, which account for the glyph structure. The additional bits appear to encode noise-related information. However, the ILM is unable to find a compositional structure in the noise, and so the additional bits are not stable. This is demonstrated in Fig. 4 where it is observed that the bits that vary between different variant images of the same glyph are also the bits that vary across generations. A shared language has arisen that can be used to communicate the identity of each glyph in a way that is stable. The agents also develop the ability to represent the perturbations that these glyphs are subjected to but this feature of the representation is stubbornly idiosyncratic and must be learned anew by each generation.

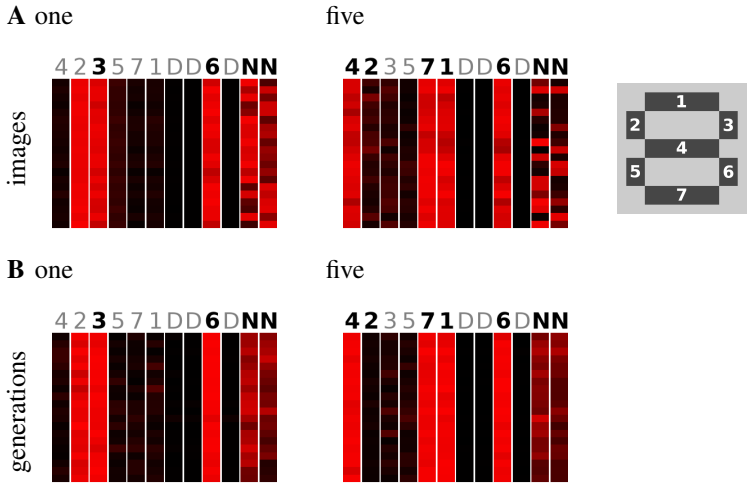

Figure 4. **Latent activations**. These show latent activations for two glyphs. In every case the columns correspond to latent nodes. Some of the columns have been labelled according to which glyph segment they respond to; the active segments are labelled in bold. For **A**, generation 60 is considered, at which point $x = 1.0$ and $c = 1.0$. For each glyph, each row depicts how one of 20 noisy '1' or '5' images are mapped to latent activations using the encoder E. No discretization is applied, though latent activations are frequently close to either zero or one. Many remain the same across variant images; these, presumably, correspond to the compositional coding of glyph segments. For **B** generations 50 to 69 are considered, for which $s = 0.496$. Each row corresponding to the latent values for the base glyph at a different generation, lower rows correspond to later generations. The same columns that show variation here are the ones that show variation across images in **A**. For these simulations a $16 \times 14 \times 12$ E_w and the noise 3 data were used; this large latent space helps the network learn quickly, but it appears some of the latent nodes, labelled 'D', cease to be active. Of the unlabelled columns, only the last two, labelled 'N', show activity in representing the noise. In this example only the second latent node codes for a glyph segment 'backwards', with low activation representing the presence of segment 2.

4. Discussion

The ILM is a very simple model, but in the semi-supervised form employed here it suggests a striking hypothesis about the origin of language: rather than being wholly concerned with encoding and decoding external utterances from a shared language, the agent must also rely on autoencoder-driven learning of its own "language of thought" (Vygotsky, 1986; Carruthers, 2002). Clearly, there is a substantial gap between the simplicity of the model and the subtlety of the language of thought idea; to bridge this, the model needs to be applied to more complex examples. This paper is a first step in this programme and comes with an interesting lesson: the same autoencoder mechanism that is useful for language learning and crucial for language evolution is also useful in learning the more peripheral parts of the pathway, analogous to primary visual processing.

5. Acknowledgements

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